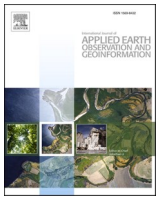
International Journal of Applied Earth Observation and Geoinformation 133 (2024) 104084

Contents lists available at ScienceDirect

International Journal of Applied Earth Observation and Geoinformation

journal homepage: www.elsevier.com/locate/jag



A refined edge-aware convolutional neural networks for agricultural parcel delineation

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ARTICLE INFO

Keywords:
Agricultural parcels
Convolutional neural networks
Edge detection
Deep supervision
Boundary refinement

ABSTRACT

Accurate delineation of agricultural parcels is crucial for ensuring food security and implementing precision agriculture practices. However, one of the primary challenges in parcel delineation revolves around the accurate extraction of parcel edges due to the diverse morphologies of these parcels. Current deep learning models for agricultural parcel delineation often produce unclosed or incomplete edges, highlighting the need to enhance the edge-awareness of the model. In this study, we proposed a refined edge-aware U-Net (REAUNet) to extract accurate agricultural parcel boundaries from Sentinel-2 images and GF-2 images. REAUNet integrates four innovative components for edge enhancement, i.e., the deep supervision, the edge detection block, the dual attention block and the refine module. The edge detection block and the dual attention block strengthen the ability of capturing spatial contextual information, and the deep supervision constrains parcel delineation across multiple scales, and the refine module further refines the parcel edges. Together, these components enhance the edge-awareness of the model, resulting in more accurate agricultural parcel delineation. From the ablation experiment, REAUNet with all four components reached the higher accuracy than the scenarios that only three out of four components were retained, which suggested that the inclusion of all four components was essential. The comparative evaluation showed that our proposed model surpassed the representative deep learning models for both thematic and geomantic accuracy. REAUNet demonstrated an increase of 2.0 % in F1-score and a decrease of 3.4 % in GTC (Global Total-Classification) compared to the SEANet, and exhibited a 0.8 % increase in IoU and a 2 % decrease in GTC relative to the MPSPNet. We also assessed the capability of the REAUNet model transferred for a new geographic region, and reported that a pre-train REAUNet model with only a small number of local samples could achieve a satisfactory accuracy. To conclude, the newly proposed REAUNet model reaches a high mapping accuracy for delineating agricultural parcels from the high-resolution images. With the powerful edge-aware capability, REAUNet can effectively utilize limited training samples to transfer a deep learning model for a new region, facilitating the generation of agricultural parcel products and a large-scale agricultural parcel mapping in practical applications.

1. Introduction

With global population growth and climate change, food security has become a significant concern worldwide (Godfray et al., 2010; Lu et al., 2015). Numerous efforts were made to address various facets of this challenge, including field yield enhancement (Garibaldi et al., 2016; Pywell et al., 2015), climate adaptation and risk management (Kuang et al., 2022), cropping intensity monitoring (Huang et al., 2024), industrialized agriculture expansion (Liu et al., 2021a). The success of these efforts is contingent on the availability of highly accurate geographic information on agricultural parcels (Potapov et al., 2022; Zabel et al., 2019). Particularly, parcel boundary maps could deliver fine-scale information on the size, geometry and spatial locations of agricultural parcel, potentially providing key inputs to the applications such as parcel size analysis (Estes et al., 2022), crop type mapping (Cai et al., 2018), crop yield forecasting (Lobell et al., 2015) and natural

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https://doi.org/10.1016/j.jag.2024.104084
Received 14 April 2024; Received in revised form 2 August 2024; Accepted 5 August 2024
Available online 16 August 2024
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disaster monitoring (Ye et al., 2021).

While enormous efforts have been made to map agricultural parcel using satellite-based technology, few datasets are available on agricultural parcel boundaries. The major impediment is that parcel boundaries tend to intergrade with surrounding vegetation from optical satellite images, leading to an algorithmic challenge for boundary detection. The edge detector was widely applied to delineate parcel boundaries by identifying object edges as discontinuities of image brightness (Crommelinck et al., 2017; Graesser and Ramankutty, 2017; Turker and Kok, 2013; Yan and Roy, 2014), but often failed to generate satisfactory segmentation results (Waldner and Diakogiannis, 2020). An edge detector computes an image gradient using an image filtering/kernel. The gradient threshold required by the edge detector for identification is universally unavailable for the whole image. Besides, the parcel sizes often vary from 0.01 to 100 ha (Waldner and Diakogiannis, 2020), which poses a knotty challenge on detecting multiscale boundaries. A delineation of a smaller parcel is more prone to noise against a larger parcel. As a result, we are often unable to apply one edge detector with a gradient threshold for parcel segmentation (Crommelinck et al., 2017; Hong et al., 2021).

To enable a more accurate delineation of crop boundary and extent, Convolutional Neural Network (CNN) has been widely applied to map agricultural parcel recently (Persello et al., 2019; Taravat et al., 2021; Waldner and Diakogiannis, 2020; Zhang et al., 2020). CNN is capable of learning complex hierarchical contextual features (Yuan et al., 2021), thereby mapping accurate parcels for various crop types in multiple scales. Among the various CNN algorithms, they can be broadly categorized into three types: model refinement, multi-modal methods, and multi-task learning methods. Model refinement primarily aims to enhance model performance by modifying the model architecture. Zhang et al. (2020) demonstrated MPSPNet to combine low-level and high-level information. The low-level information captures edge details related to object boundaries, while the high-level information learns long-range spatial dependencies, both of which aid in the recognition of agricultural parcels. Waldner and Diakogiannis (2020) proposed ResUNet-a, which incorporates parallel branching atrous convolutions. This enables multi-scale feature extraction and fusion across different receptive fields. Pan et al. (2023) proposed the E2EVAP framework to extract the vertices of each parcel boundary individually, where semantic-contour interaction and topological loss were used to aggregate foreground features and establish topological relationships. This approach has multiple directions for model improvement, allowing the model to focus on different kinds of situations. Multi-modal methods use multiple data as complementary information to improve parcel extraction accuracy. Cai et al. (2023a) proposed DSTFNet, which integrates high-resolution images with medium-resolution time-series images. This model captures spectral dynamics, aiding in the extraction of diverse agricultural parcels. Sun et al. (2022) combined multi-temporal optical and SAR remote sensing data to address the limitations of optical data availability, especially in cloudy and rainy region. This approach compensates for the shortcomings of using a single data source. However, it can present challenges related to data collection and availability. Multi-task learning methods enable the model to simultaneously extract correlated information, such as parcel extent and parcel edges. Long et al. (2022) proposed BsiNet with multi-task learning and attention mechanism to better delineate agricultural parcels. Wang et al. (2023) introduced BSNet, which employs multitask learning to extract agricultural parcels and their edges, and then fuses these results to produce the final output. However, multi-task learning is a training approach that does not alter the model structure and does not inherently enhance the model ability to better focus on edges.

Despite their impressive accuracy, current deep learning models still face limitations in delineating agricultural parcels with varying morphologies and complex shapes (Lesiv et al., 2019; Zhang et al., 2021). Many CNN-based approaches might lose edge information during the backpropagation process, causing vanishing gradient issue. To tackle this issue, Li et al. (2023) presented an edge-aware SEANet incorporated local and global edge semantics at multiple levels to derive more accurate agricultural parcel extraction. While SEANet demonstrated efficacy in delineating agricultural parcels of diverse shapes and sizes, the method exhibited certain limitations. Primarily, the proposed edge-aware approach generates edge feature representations by using 1×1 convolutions from the scratch which requires considerable training samples to achieve an acceptable accuracy. Additionally, the extracted agricultural parcel boundaries were either indistinct or obscured by surrounding land covers (Li et al., 2023), suggesting more edge refinement needed on the segmentation results.

In this study, our aim was to delineate agricultural parcels with boundaries that are both precise and geometrically sound, employing a deep learning model integrated with refined edge-aware methods. We developed a sematic segmentation neural network, named Refined Edge-Aware U-Net (REAUNet), to enable an extraction of multi-scale parcel information and fine-tune parcel edge. REAUNet contains four innovative components to enhance its ability of delineating parcel boundaries: the deep supervision, the edge detection block, the dual attention block and the refine module. Different with the previous work such as SEANet, REAUNet directly integrates traditional edge detection operators instead of data-driven tuning, which allows the model to diminish the reliance on extensive labeled dataset and to reduce the required volume of local samples for spatial generalization. Finally, the model performance will be assessed in comparison to other state-of-the-art deep learning models. The main contributions are:

(1) developing a refined edge-aware deep learning model to effectively delineate agricultural parcels from remote sensing images;
(2) formulating a post-processing workflow to optimize the geometry and shape of extracted parcels;
(3) demonstrating the capability for utilizing limited samples to transfer a trained model for a new region.

2. Study areas and datasets

2.1. Study areas

Four testing sites were selected to evaluate our method (Fig. 1): the Netherlands (NL), Denmark (DK), and two regions Huantai (HT) and Tarim River (TR) in China. Two image sources were tested, i.e., 10-m Sentinel-2 and 1-m Gaofen-2 (GF-2) images. Sentinel-2 images were acquired from Google Earth Engine (GEE) for the areas NL and DK, which were preprocessed through atmospheric correction, radiometric correction, geometric correction, and terrain correction. We chose data for the year 2022 with cloud cover less than 10 %. A temporal compositing technique was employed to generate composite data covering the entire study area based on the mean value. The reference data for agricultural parcels in NL was available from the Basisregistratie Gewaspercelen (BRP) (https://www.pdok.nl/introductie/-/article/basisregistratie-gewaspercelen-brp-), while corresponding data in DK was sourced from the European Union Land Parcel Identification System (LPIS) (https://collections.eurodatacube.com/). The polygons of agricultural parcels in BRP and LPIS were manually created by the agencies. The other two study regions in China comprised of HT in Shandong Province, and TR in Xinjiang Province. High-resolution GF-2 images were obtained respectively on March 2022 for HT, and on September 2022 for TR. The preprocessing steps for GF-2 imagery included atmospheric correction, radiometric calibration, geometric correction, and PAN sharpening, resulting in the final 1-m resolution multispectral data. The ground truth parcels for GF-2 imagery were crafted through manual annotation.

The four regions cover a diversity of agricultural systems. In the NL area, agricultural parcels are diverse, including well-organized large-scale parcels, intensive narrow parcels, and irregular parcels. The main crops are ornamental crops, potatoes, sugar beet, maize, cereals, and

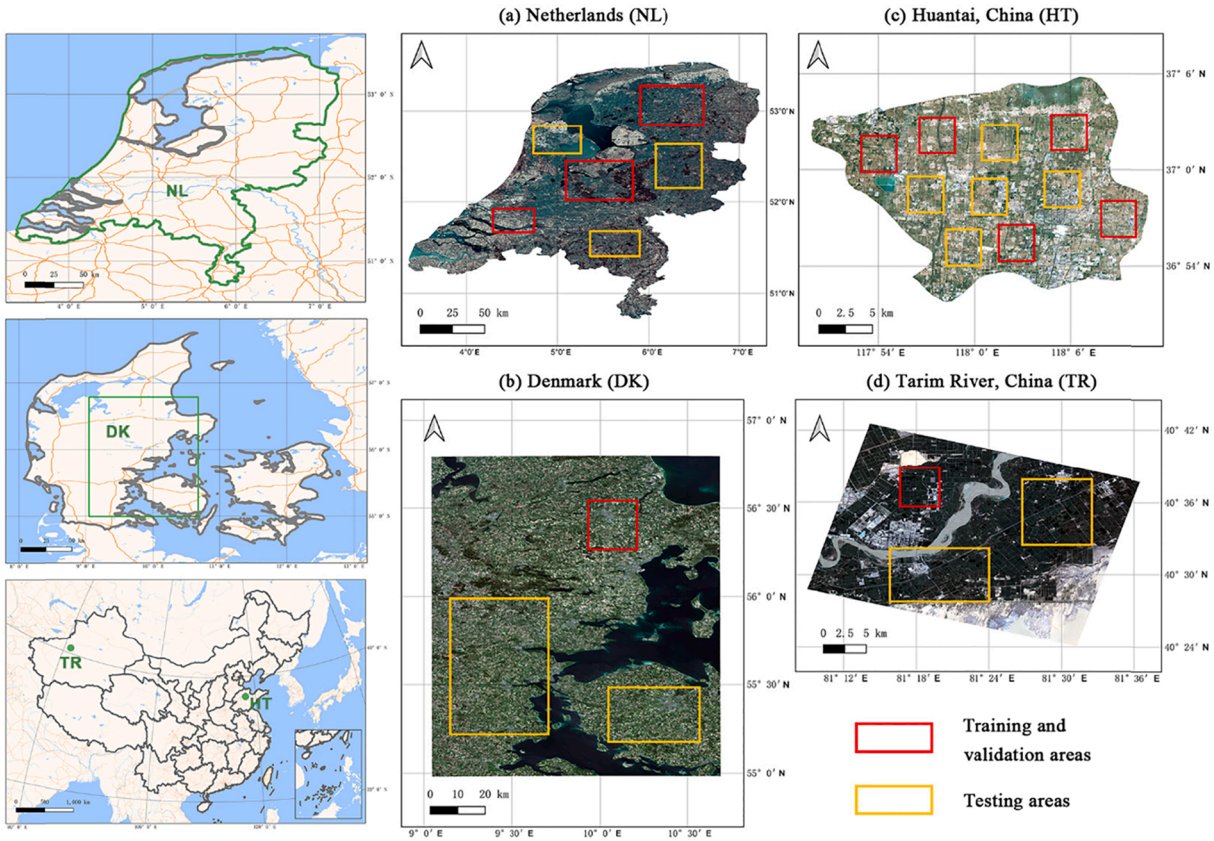

Fig. 1. Overview of four study areas including (a) Netherlands (NL), (b) Denmark (DK), (c) Huantai (HT) and (d) Tarim River (TR). Sentinel-2 images were selected for (a) NL and (b) DK, and GF-2 images were selected for (c) HT and (d) TR. The data from (b) DK and (d) TR were exclusively used in the transfer experiment. The red boxes denote the data utilized for training and validation, while the yellow boxes represent the test data. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

onions. While similar to NL with well-organized agricultural parcels, DK also includes smaller, more irregularly shaped parcels due to its diversified terrain, with barley, wheat, rapeseed, and corn as the primary crops. These two areas require the model to delineate different patterns and complex boundaries of agricultural parcels. In the HT area, agricultural parcels vary in size and shape, predominantly cultivated for wheat and corn with parcels at different sizes. The XJ area features vast land resources with large agricultural parcels mixed with smaller, irregular ones. The differences among the four areas highlight the challenge of calling for model's versatility and generalizability across various landscapes.

2.2. Datasets

The training, validation, and testing regions were randomly selected from the images, as displayed in Fig. 1. The regions designated for training and validation were cropped into chips with a uniform size of 256*256 pixels and subsequently divided into training and validation sets using a random 8/2 split. During the training process, a total of 2,287 training chips and 572 validation chips were used in the NL dataset. The HT dataset consisted of 1,962 training chips and 491 validation chips. We employed data augmentation to enhance the amount and diversity of training samples. For each sample, one of the five transformations, i.e., unchanged, horizontally flipped, vertically flipped, rotated 90 degrees, and rotated 180 degrees, was randomly applied. In the transfer experiment, we selected 100 training chips from each new region. It is noteworthy that we used all pixels in the testing area ("yellow rectangle" in Fig. 1) for the final accuracy assessment, instead of selected chips.

3. Methods

The workflow of our study is illustrated in Fig. 2. After image preprocessing, we will train the new Refined Edge-Aware U-Net (REAUNet) model which consisted of four innovative components for an improved agricultural parcel identification, then apply a post-processing step to enhance edge delineation, and finally perform a multi-angle accuracy assessment.

3.1. Refined edge-aware U-Net (REAUNet)

We developed the architecture of our proposed REAUNet based upon U-Net (Ronneberger et al., 2015) with four innovative components (Fig. 3). The proposed architecture adopts the ResBlock-34 (He et al., 2016) as the basic convolutional block (Fig. 3b), and uses 5 × 5 convolutional kernels in some of the convolutional layers of the deep encoder to expand the receptive field and better capture global information. The model can be divided into two parts: the predict module and the refine module, with the prediction module serving as the model backbone. This backbone employs an encoder-decoder structure with skip connections, enabling the simultaneous capture of high-level global contexts and low-level detailed features. It is organized into four symmetric stages and a bridge stage, incorporating additional convolutional layers at the deeper stages. In each stage of the encoder, an edge detection block is inserted between the convolutional blocks, and a dual attention block is added at the end. The bridge stage utilizes a pyramid pooling module for multi-scale feature fusion. It combines features from multiple scales, thereby generating global context knowledge that encompasses both low-level and high-level information (Zhao et al., 2017). In lieu of maximum pooling, a convolution operation with the kernel

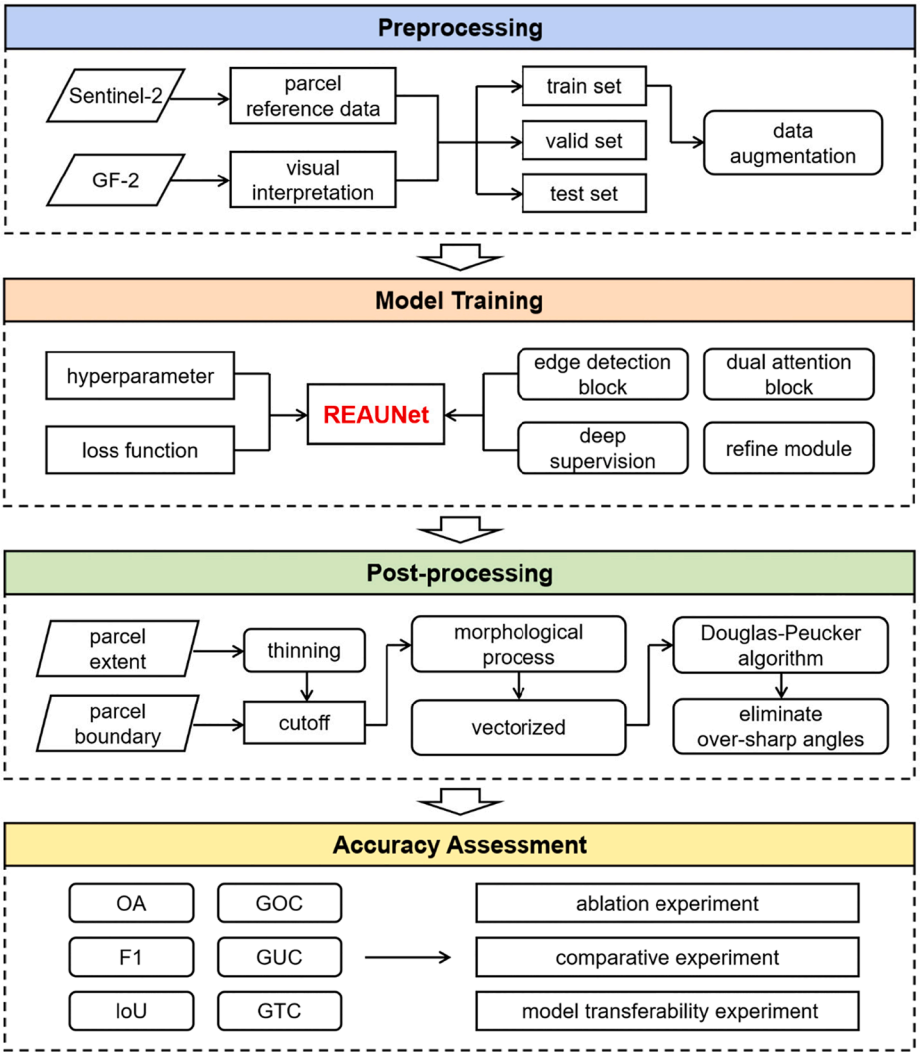

Fig. 2. The workflow of the study.

size of 3 × 3 and the stride of 2 is used for the down-sampling process to minimize information loss, while a bilinear interpolation is employed for the up-sampling process. The below are the details for four innovative components within REAUNet.

3.1.1. *Deep supervision*

Deep supervision is a widely employed technique in deep learning models for edge detection (Liu et al., 2017; Xie and Tu, 2015). This approach involves adding supervision signals at hidden layers of a neural network, effectively guiding the learning process at multiple stages. Deep supervision could alleviate the problem of gradient vanishing and provide robust regularization of learned features, which is particularly beneficial for a case of small training data and shallower networks. In our proposed REAUNet, deep supervision is integrated by combining multiple coarse outputs from the predict module with the refine output from the refine module, which ensures that supervision is applied across different-sized feature maps. By supervising multiple layers, the network can enhance feature regularization and learn richer feature representations.

3.1.2. *Edge detection block*

The edge detection block (Fig. 3c) was inserted into the middle of each stage to integrate edge gradient information during the feature extraction step, so that we could reinforce the ability of recognizing agricultural parcel edges. This block applies four different Sobel filters to detect gradients in the feature maps along four directions (Ranjan and Avasthi, 2023), and subsequently fuses the features in each gradient direction through a convolutional layer. Common deep learning models often lack expert knowledge and rely heavily on data-driven learning. REAUNet augments the model with traditional edge detection operators. By leveraging this empirical knowledge, REAUNet bolsters the model capability to extract sophisticated semantic features from limited samples. Also, the edge detection block compensates for omissions of edge features through residual connections.

3.1.3. *Dual attention block*

The dual attention block (Fig. 3d) integrates channel attention and spatial attention. Each component performs parallel computations of global average pooling and global maximum pooling, inspired by previous works (Roy et al., 2018; Woo et al., 2018). The benefit of the proposed attention mechanism is that it explores the spatial-contextual relationship among pixels in addition to interaction among feature channels. This block enables the model to better focus on areas that are prone to errors, such as the edges of agricultural parcels.

3.1.4. *Refine module*

The refine module, inspired by BASNet (Boundary-Aware Salient Network) (Qin et al., 2019), was implemented to further enhance edge prediction accuracy of the predict module (Fig. 3a). The predict module generates a coarse map with commendable regional accuracy, yet there

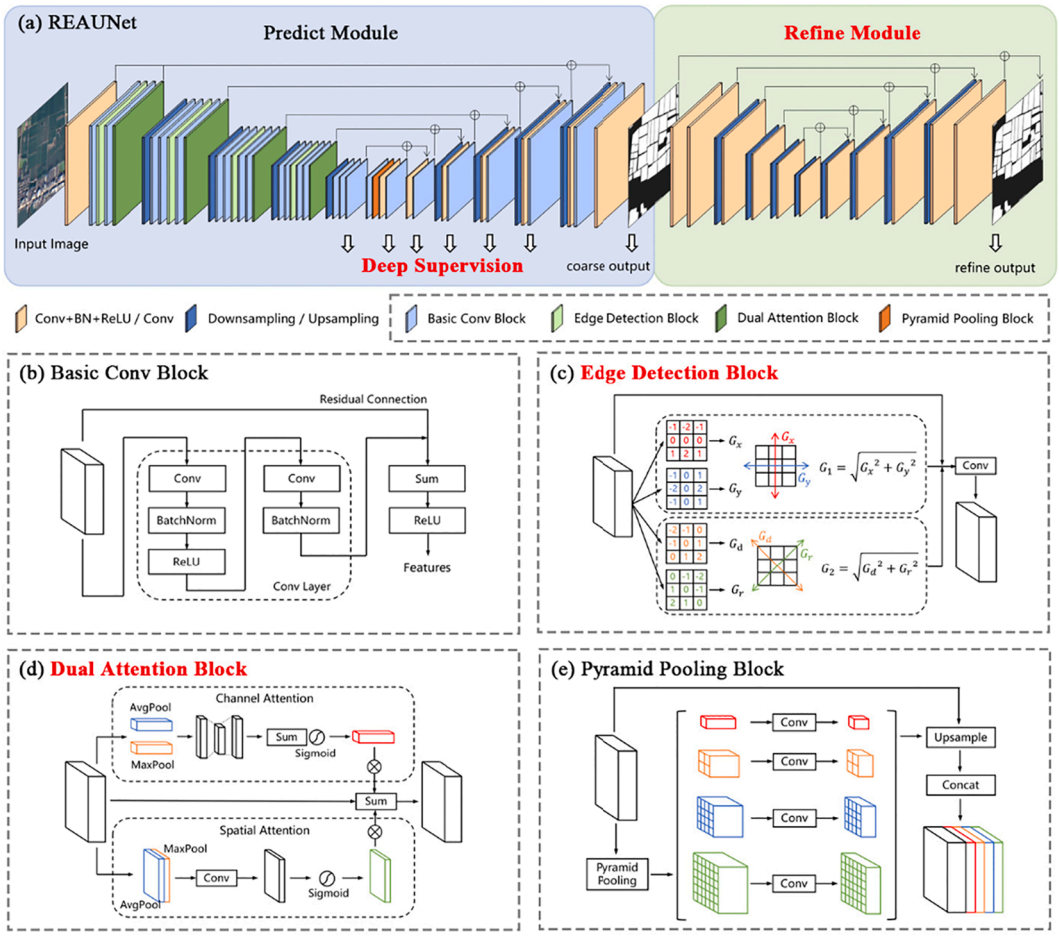

Fig. 3. The architecture of the proposed REAUNet for agricultural parcel delineation. Four innovative components are highlighted as the red color: deep supervision, edge detection block, dual attention block, and refined module. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

remains potential for enhancing edge quality. The refine module serves to refine the segmentation results by rectifying omission/commissions in the coarse map. It operates a straightforward residual encoder-decoder architecture to learn the residual map between the output of the predict module and the ground truth. Both the encoder and decoder consist of four layers symmetrically, each containing only one convolutional layer.

Finally, we utilized labels for the agricultural parcel extent and the parcel boundary to construct two models respectively for agricultural parcel and boundaries, and then derived the ultimate parcel delineation result by merging the extent mask and the boundary mask.

The codes for the REAUNet have been made publicly available in https://github.com/Remote-Sensing-of-Land-Resource-Lab/REAUNet.

3.2. Loss functions and training details

Different loss functions are employed for parcel extent training and parcel boundary training. The segmentation of agricultural parcel extent can be viewed as a semantic segmentation task. The loss function for parcel extent combines a binary cross-entropy (BCE) loss L_{BCE} and a dice loss L_{Dice} to tackle sample imbalance and ensure the stability of the training process. This can be defined as:

$$L_{extent} = 0.5 \times L_{BCE} + L_{Dice} \tag{1}$$

$$L_{BCE} = -\frac{1}{N}\left(\sum_{i=1}^{N}(y_i \cdot \log\widehat{y}_i + (1-y_i)\cdot\log(1-\widehat{y}_i))\right) \tag{2}$$

$$L_{Dice} = 1 - \frac{1}{N}\sum_{i=1}^{N}\frac{2y_i\cdot\widehat{y}_i}{y_i+\widehat{y}_i} \tag{3}$$

where y_i denotes the binary ground truth label, $\widehat{y}_i$ represents the predicted result, and N signifies the total number of pixels.

Parcel boundary delineation is akin to an edge detection task, and the distribution of edge pixels and non-edge pixels is typically highly imbalanced. Accordingly, we employed a dynamically weighted BCE loss, where the weights adapt to the proportion of positive and negative pixels. In addition, we integrated a sum-based loss calculation mode instead of averaging, thereby amplifying the loss weights for the limited number of edge samples. The boundary loss function was defined as:

$$L_{edge} = \sum_{i=1}^{N} w\cdot(y_i\cdot\log\widehat{y}_i + (1-y_i)\cdot\log(1-\widehat{y}_i)) \tag{4}$$

$$w = \begin{cases} \dfrac{N_{negative}}{N_{batch}} & if\ y_i = 1 \\ \dfrac{\alpha \times N_{positive}}{N_{batch}} & if\ y_i = 0 \end{cases} \tag{5}$$

where w refers to the weights calculated dynamically based on the number of positive and negative samples in each batch, and α is the sample balance coefficient set to 1.1.

Our hyperparameter tuning process focuses on two critical parameters: the learning rate and the weight decay. The learning rate selection method, referred to Smith (2017), involved setting a range from 10^{-5} to 10^{-2}. We conducted 8 epochs of training with various learning rates and

then chose the learning rate associated with the highest final accuracy as the initial learning rate. Fig. 4(a) indicates that the learning rate of 3×10^{-4} achieved the highest accuracy, which was considered the optimal initial learning rate. After determining the initial learning rate, we performed an experiment comprising 100 training epochs, varying three weight decay parameters (10^{-3}, 10^{-4}, 10^{-5}). Fig. 4(b) shows that the weight decay of 10^{-4} achieved the lowest loss value during training. As a result, we employed the Adam optimizer with an initial learning rate of 3×10^{-4} and a weight decay of 1×10^{-4}. The models underwent training for a maximum of 200 epochs with a batch size of 8. In cases of the model loss plateauing for 30 consecutive epochs, the learning rate was reduced by a factor of 0.1. If the validation set loss showed no improvement for 50 consecutive epochs, we implemented an early stopping to mitigate overfitting.

3.3. Post-processing

We designed a post-processing workflow to further improve the edge delineation. The specific steps were outlined as follows:

(1) applied a 1-pixel buffer to the parcel boundary results, reducing the boundary interruption in the results;
(2) performed a morphological thinning operation on the boundary results corresponding to the previous buffer;
(3) utilized the parcel extent mask and the boundary mask to delineate individual agricultural parcels with the cutoff method, and the cutoff threshold was set to 0.5 (Waldner and Diakogiannis, 2020);
(4) conducted morphological processing on the binary results to eliminate holes and small isolated pixels;
(5) vectorized the results and employed the Douglas-Peucker algorithm to remove redundant nodes, optimizing the edges of agricultural parcels;
(6) detected and eliminated over-sharp angles.

Fig. 5 exemplifies the improvement brought by the post-processing step. We applied a morphological thinning operation to narrow the boundary map and subsequently utilized the cutoff method to isolate individual agricultural parcels. Although the deep learning model successfully captured the boundaries of agricultural parcels, some breakpoints persisted in the results, leading to irregularly shaped parcels being obtained, as shown in Fig. 5(c). To address this, the Douglas-Peucker algorithm and the over-sharp angle elimination algorithm were employed to optimize the spatial geometric shapes of the parcels.

3.4. Comparison with representative CNN models

In order to evaluate the effectiveness of our method, we conducted a comparative analysis of eight CNN models specifically designed for segmentation tasks and agricultural parcel delineation. Table 1 presents the brief descriptions for these CNN models.

3.5. Model transferability experiment

To assess the capability of REAUNet being applied to different agricultural areas, we conducted a model transferability experiment. We tested the model trained from NL on DK region for Sentinel-2 images, and the model from HT on the TR region for GF-2 images. In this experiment, we defined three scenarios: No Training (NT) implies that the model, previously trained in other regions, was taken to make predictions directly in the new region (e.g., using the model trained on NL data to delineate parcels in the DK area); Direct Training (DT) trained the model merely using local samples; Transfer Training (TT) pretrained a model using data from another region first, followed by continued training with local samples from the new region. The NT scenario can be also viewed as a validation of the model generalizability when confronted with a new region.

In the new region selected for the transfer experiment, we opted for a small area to generate training samples, resulting in approximately 100 samples of size 256×256 pixels. The $\text{REAUNet}_{\text{DT}}$ followed the same protocol as REAUNet, with a maximum of 200 epochs. On the other hand, $\text{REAUNet}_{\text{TT}}$ was trained for a shorter duration, specifically 20 epochs.

3.6. Accuracy assessment

We used a total of six evaluation metrics to assess the performance of model for delineating agricultural parcels from both pixel-based and object-based perspectives, i.e., Overall Accuracy (OA), Intersection over Union (IoU), F1-score (F1), Global Over-Classification (GOC), Global Under-Classification (GUC) and Global Total-Classification (GTC). The pixel-based metrics (OA, IoU and F1) were defined as follows:

$$OA = \frac{TP + TN}{TP + FN + FP + TN} \tag{6}$$

$$IoU = \frac{TP}{TP + FP + FN} \tag{7}$$

$$F1 = \frac{2{*}TP}{2{*}TP + FP + FN} \tag{8}$$

where TP, TN, FP, and FN refer to True Positive, True Negative, False

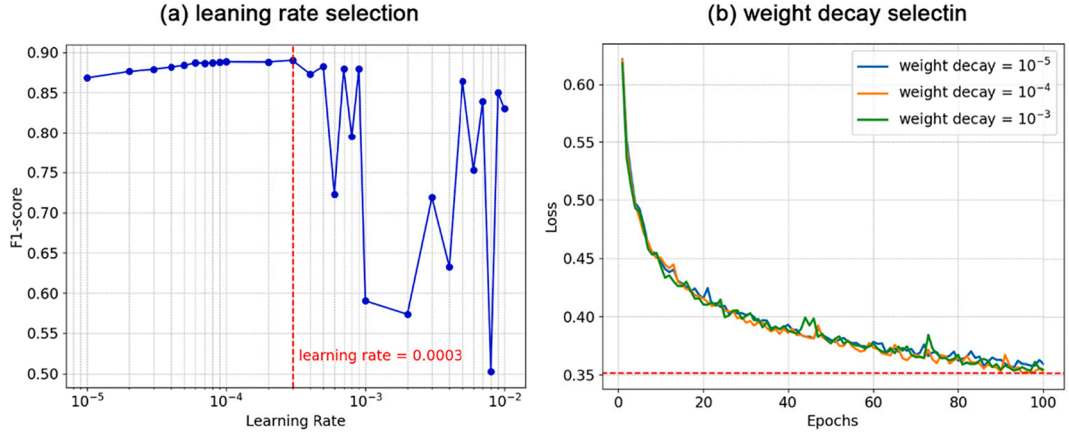

Fig. 4. Hyperparameter selection. (a) The highest accuracy was achieved at a learning rate of 3×10^{-4}. (b) The lowest loss value was obtained with a weight decay of 10^{-4}.

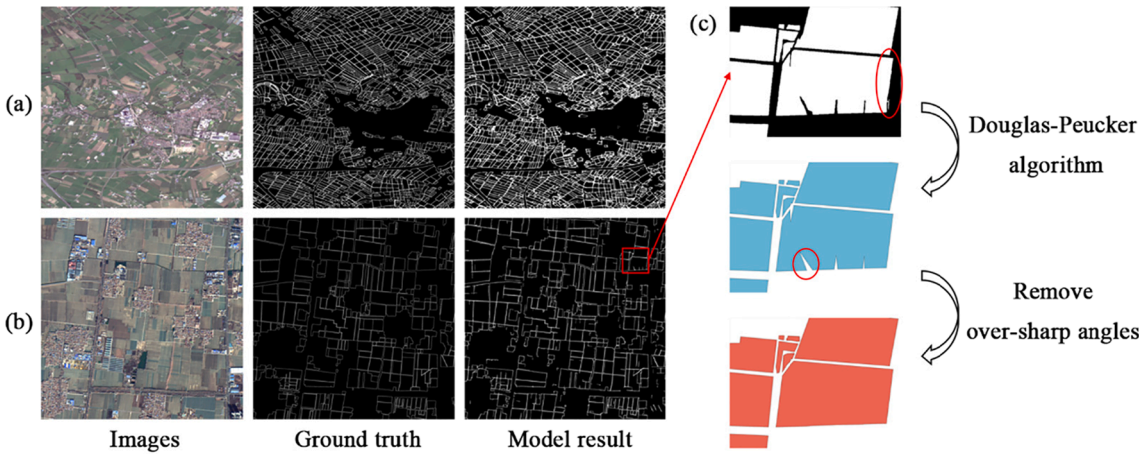

Fig. 5. Model results of parcel boundaries. (a) represents area NL, (b) represents area HT, and (c) shows the refined parcels formed by the post-processing method.

Positive, and False Negative, respectively.

To better assess spatial accuracy of extracted parcels, we adopted three object-based accuracy metrics as a complementary evaluation, based upon under-segmentation and over-segmentation errors (Li et al., 2023; Ye et al., 2018), which were defined as followed:

$$GOC = \sum_{i=1}^{m}\left(OC(S_i) \times \frac{area(S_i)}{\sum_{i=1}^{m} area(S_i)}\right) \tag{9}$$

$$GUC = \sum_{i=1}^{m}\left(UC(S_i) \times \frac{area(S_i)}{\sum_{i=1}^{m} area(S_i)}\right) \tag{10}$$

$$GTC = \sum_{i=1}^{m}\left(TC(S_i) \times \frac{area(S_i)}{\sum_{i=1}^{m} area(S_i)}\right) \tag{11}$$

where $S_{i,i=1,2,\cdots,m}$ denotes the classified parcels in model results. $OC(S_i)$, $UC(S_i)$ and $TC(S_i)$ were defined as:

$$OC(S_i) = 1 - \frac{area(S_i \cap O_i)}{area(O_i)} \tag{12}$$

$$UC(S_i) = 1 - \frac{area(S_i \cap O_i)}{area(S_i)} \tag{13}$$

$$TC(S_i) = \sqrt{\frac{OC(S_i)^2 + UC(S_i)^2}{2}} \tag{14}$$

Where $O_{i,i=1,2,\cdots,n}$ represents the reference objects in the ground truth, which is the parcel closest in distance to S_i.

We used these accuracy metrics (Eq. 6–11) for an ablation experiment and the final accuracy report. The ablation experiment assessed the accuracy loss after removing each component out of the four innovative components for the REAUNet, so that we could understand the accuracy contribution of individual component.

4. Results

4.1. Ablation experiment

Table 2 depicts the accuracy for different settings of REAUNet in the ablation experiment. In comparison to the original REAUNet, all other model settings exhibited notably higher GUC/GTC, indicating the four components all contribute to the final geometric accuracy. Fig. 6 showcases that the REAUNet with all four components (the third column "REAUNet") significantly presents the least false negative errors at the edges of parcels. Among the four modules, the removal of deep supervision had the most pronounced impact on accuracy, highlighting the crucial role of multi-scale deep supervision in the accurate delineation of agricultural parcels.

To further explore the decision process of REAUNet and enhance the model transparency, we outputted the feature maps of the model at different stages to show how the input images is transformed passing through the layers. Fig. 7(b) illustrates the feature maps from the shallow and deep layers for the encoder stage, where the bright pixels indicate the patterns the REAUNet is seeking for. Shallow features (Encoder #1) trains most of the information present in the image. Deep features (Encoder #4), on the other hand, are high-level features for seeking an abstract representation. Some hidden class information, such as the impervious surface (the fourth column), become recognizable. The output feature maps for the decoder present a reversing trend of increasing details (Fig. 7c) from Decoder #1 to Decoder #5. Benefited from the design of deep supervision, REAUNet will integrate these feature maps generated from different scales and guide these intermediate layers towards the desired solution, enabling the model to better capture hierarchical relationships among parcel features.

4.2. Comparative experiment

The accuracy assessment results for REAUNet and other deep learning models were presented in Table 3. The REAUNet model consistently outperformed the other models for most accuracy metrics: in the NL study area, REAUNet demonstrated an increase of 2.0 % in F1-score and a decrease of 3.4 % in GTC compared to the second-best SEANet; in the HT study area, REAUNet exhibited a 0.8 % increase in IoU and a 2 % decrease in GTC relative to the second-best model, MPSPNet. Fig. 8 visualizes the agricultural parcel delineation maps from various deep learning methods. In contrast to other models, our proposed method notably reached the lowest false negative errors at parcel edges. Additionally, REAUNet proves effective in minimizing false negative errors (the blue region) in certain regions of the HT study area, particularly where parcel boundaries are less evident. Thanks to its powerful edge-awareness, REAUNet achieved excellent extraction results in study areas with different data sources. It successfully extracted complex and diverse agricultural parcels in the NL region and delineated parcels of varying sizes in the HT region. The strength of REAUNet lies in performing complex and diverse parcel extraction at multiple scales, and its delineation results in these regions surpass those of several representative models.

4.3. Agricultural parcel delineation results

Fig. 9 displays the extraction results of REAUNet for the NL and HT areas, showcasing the successful mapping of agricultural parcels from

Table 1
The representative deep learning methods for agricultural parcel delineation.

Method	Model description	Reference
U-Net	• U-Net consists of a contracting path to capture context and a symmetric expanding path that enables precise localization; • Every step in the expansive path is followed by a concatenation with the correspondingly cropped feature map from the contracting path.	(Ronneberger et al., 2015) (Kumar and Jayagopal, 2021)
RCF	• RCF is an end-to-end convolutional network designed to extract edges; • RCF keeps five convolutional stages in VGG16, and the feature maps from the same stages are added up then upsampled to generate an edge probability map; • A fusion output is obtained by concatenating all 5 edge probability maps in channel dimension.	(Liu et al., 2017) (Liu et al., 2019)
MPSPNet	• MPSPNet incorporates three core structures: ResNet with dilated convolution, spatial pyramid average pooling module, and fusion of low-level and high-level information.	(Zhang et al., 2020)
ResUNet-a	• ResUNet-a consists of residual units with multiple parallel branches of atrous convolutions, each with a different dilation rate to increase the receptive field; • ResUNet-a produces four output layers: the extent mask, the boundaries, the distance mask, and full reconstruction of the input image in the HSV color.	(Diakogiannis et al., 2020) (Waldner and Diakogiannis, 2020)
BsiNet	• BsiNets uses one single decoder to produce three types of information mask, boundary, and distance; • SGE attention module is incorporated into BsiNet to extract enhanced features, utilising the global information and scaling the importance of different channels.	(Long et al., 2022)
FRNet	• FRNet is divided into four parallel streams in the horizontal direction, i.e., the first stream with full resolution and the additional three with resolution decreased by half; • Different parallel streams are connected and fused by the multi-resolution fusion unit (MRFU).	(Xia et al., 2022)
RAUNet	• RAUNet uses ResNet to capture higher-level abstract features in the encoder; • RAUNet integrates coarse- and fine-scale features to suppress irrelevant noise and highlight salient regions with a gated attention mechanis in the decoder.	(Cai et al., 2023b)
SEANet	• SEANet consists of two main components corresponding to PRoI and PEoI feature extraction; • SEANet constructs a stack of convolution blocks to derive edge feature maps at multiple levels.	(Li et al., 2023)

two distinct remote sensing image sources, Sentinel-2 and GF-2. The vectorized agricultural parcel boundaries in Fig. 9(c) and (d) revealed that REAUNet accurately delineates large, well-organized agricultural parcels, as well as dense, elongated parcels in the NL region. Fig. 9(e) and (f) displayed the results for agricultural parcels with complex shapes. Likewise, in the HT region, the model successfully identified various agricultural parcels of different shapes and sizes.

4.4. Model transferability experiment

Table 4 presents the accuracy assessment for the three scenarios in the transferability experiment. The $REAUNet_{TT}$ achieved the highest accuracy and surpassed the model trained from scratch for 200 epochs by simply continuing training for 20 epochs using only 100 local samples. Fig. 10 shows the resultant maps for these three scenarios. $REAUNet_{NT}$ achieved the lowest accuracy and exhibited numerous false negative pixels ("the blue region"), indicating that the model struggled in these regions without local samples. Moreover, $REAUNet_{TT}$ presented a higher accuracy than $REAUNet_{DT}$ for both DK and TR, which suggested the superiority of REAUNet in transferring model for a new region. The transfer trained model incurred a training cost of less than 5 min and exhibited a notable 10–15 % improvement in F1 over the original model.

5. Discussion

5.1. Comparison to representative models

Our proposed REAUNet outperformed all representative models for agricultural parcel extraction that were selected. There were mainly four innovations to specifically improve model performance, and the results in the ablation experiment indicated that all four components positively contribute to the final accuracy of the agricultural parcels. In contrast to MPSPNet (Zhang et al., 2020), which incorporates residual dilated convolution and pyramid pooling module, our model integrates an attention mechanism and multi-scale deep supervision. The dual attention block enables the model to better tackle important regions and critical features of the parcels during the feature extraction process, enhancing its adaptability in handling parcels with diverse shapes and complex structures. Deep supervision allows the model to detect various levels of edge structures from different scales. This capability enables the capturing of edge structural information pertaining to both local details and the overall context. Notably, we observed that removing the deep supervision of the model causes the most accuracy loss in the ablation experiment. The attention mechanism and deep supervision in REAUNet empower it to effectively identify agricultural parcels of varying sizes, resulting in reduced over-segmentation compared to MPSPNet.

Furthermore, in comparison to SEANet (Li et al., 2023), which employs similar edge-aware and multi-scale segmentation techniques, we enhanced the accuracy of edge region delineation around parcels by integrating the edge detection operator into REAUNet (i.e., Edge Detection Block). This innovation leverages four different Sobel operators based upon empirical knowledge to detect feature map edges across various gradients, mitigating the impact of the morphology of agricultural parcels. Besides, to tackle the issue of incorrect pixels along parcel edges, we incorporated a small residual refine module following the main model. The demonstrated effectiveness of the refine module in the realm of salient object detection showcased its capability to efficiently reduce classification errors at target edges through residual learning

Table 2
Accuracy assessment in the ablation experiment. ED: the edge detection block. DA: the dual attention block. RM: the refine module. DS: the deep supervision.

Method	NL						HT					
	OA	F1	IoU	GOC	GUC	GTC	OA	F1	IoU	GOC	GUC	GTC
REAUNet	**0.9473**	**0.9498**	**0.8372**	**0.1363**	**0.2148**	**0.2017**	**0.9481**	**0.9454**	**0.9096**	0.1123	**0.2234**	**0.2074**
$REAUNet_{noED}$	0.9411	0.9398	0.8216	0.1502	0.2347	0.2251	0.9399	0.9425	0.8985	0.1101	0.2770	0.2296
$REAUNet_{noDA}$	0.9302	0.8925	0.7948	0.1491	0.2605	0.2360	0.9393	0.9426	0.8984	0.1189	0.2867	0.2337
$REAUNet_{noRM}$	0.9402	0.9092	0.8190	0.1605	0.2487	0.2347	0.9386	0.9416	0.8966	0.1160	0.2742	0.2370
$REAUNet_{noDS}$	0.9289	0.8877	0.7875	0.1571	0.2858	0.2549	0.9304	0.9326	0.8807	**0.1097**	0.3181	0.2701

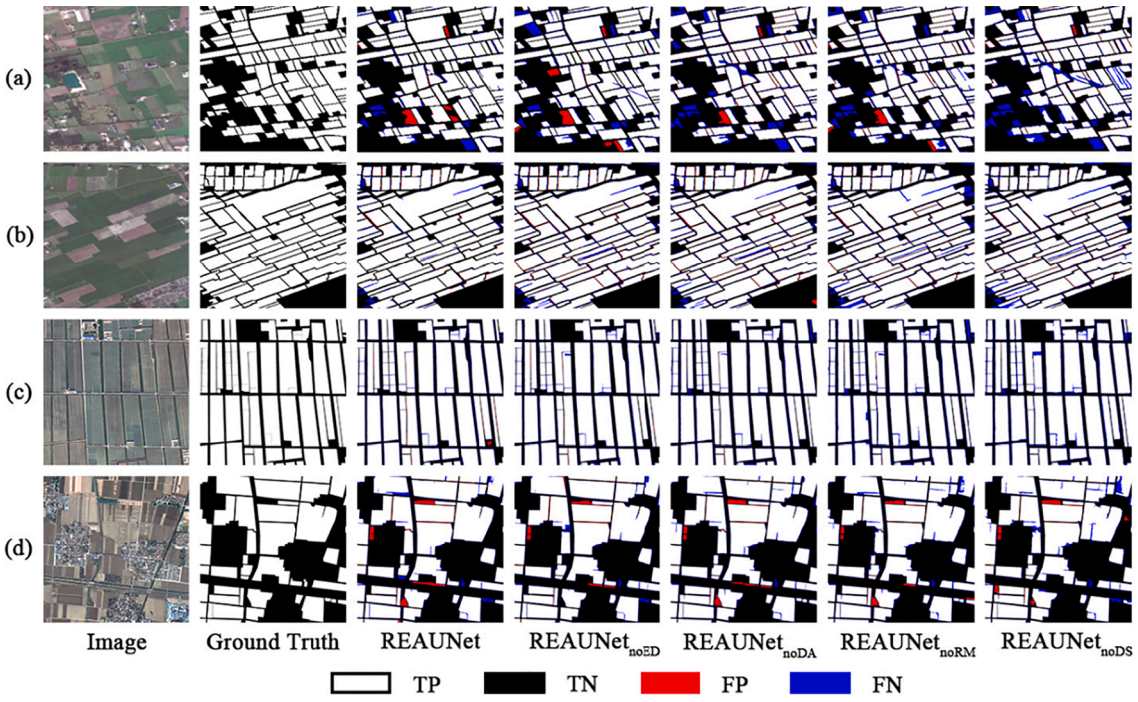

Fig. 6. Delineation results in the ablation experiment. (a) and (b) represent area NL, while (c) and (d) represent area HT.

(Qin et al., 2019). The combination of empirical knowledge and residual refinement further enhances the model ability for edge delineation, facilitates more precise identification for ambiguous crop edges, thus diminishes model errors.

We also designed a post-processing workflow to optimize the geometry shape of parcels. The presence of different crops in agricultural parcels could lead to adjacent areas displaying different colors in remote sensing images. This variation poses a challenge for the model in distinguish whether it is parcel boundaries, often resulting in erroneous short boundary lines. These errors contribute to false negatives within agricultural parcels when fusing the parcel extent and boundary results (Fig. 5c). To address this issue, we used the Douglas-Peucker algorithm and the over-sharp angles detection and elimination algorithm to mitigate these false negatives. Through this post-processing approach, our method produced agricultural parcels with a more coherent spatial geometry.

In this study, we opted for convolutional neural network (CNN) as the deep learning model for agricultural parcel delineation, a choice that has demonstrated significant success in the realm of image segmentation. However, it is noteworthy that Transformer-based methods, originally developed for natural language processing (NLP), are gaining popularity in image segmentation and have shown remarkable results in recent years (He et al., 2022; Liu et al., 2021b; Vaswani et al., 2017). The primary rationale for selecting CNN is the extensive use of linear layers and self-attention mechanisms in Transformers, resulting in a larger number of parameters and slower computation. Conversely, a crucial characteristic of CNN is weight sharing, enabling it to efficiently reduce the number of parameters when handling multidimensional matrices. CNN excels in extracting local features from images using convolutional kernels, whereas Transformer possesses the capability to capture long-range dependencies.

5.2. Model transferability

In the practical application of the deep learning model, it frequently encounters scenarios involving unknown data, placing high demands on the transfer ability of the model. However, models trained using data from non-targeted regions often fail to exhibit a sound accuracy for target regions. In the model transferability experiment, the unsatisfactory results achieved by the NT model suggested that it was challenging to obtain good performance when the trained model was applied directly to an unknown region. Agricultural parcels in various regions and during different seasons display significant variations in spatial patterns and spectral features from remote sensing images, presenting variations in terms of the shape and size of agricultural parcels. In plains or areas with mechanized farming operations, agricultural parcels tend to be large and regular. In contrast, mountainous and hilly regions often have smaller, more fragmented parcels. Additionally, the different growth stages of crops across seasons result in varying spectral features in remote sensing images as well. For instance, the greenness will show notable differences between the early and mature stages of crop growth from remote sensing images. Additionally, different crop species across the two time periods can also lead to variations in boundary delineation. This discrepancy poses difficulties for the model when confronting unknown data, which requires the inputs of the new samples representative for the geography and the mapping seasons of the images to be classified. However, redrawing labels and reconstructing the training model from the scratch could be costly. The model transferability experiment simulated three scenarios, i.e., direct model transfer, rebuilding from the scratch, and a mixture of non-local and local samples. The results highlighted the potential of REAUNet being transferred to a new region only with a small number of local samples. This was because that the proposed REAUNet leveraged multi-scale deep supervision to segment agricultural parcels across various scales, yielding commendable extraction results when dealt with parcels of different sizes in new regions. Moreover, we integrated traditional edge detectors into the model, enabling a feature extraction guided by the window techniques which was generalized from numerous past computer vision studies. The potential model transferability shown by REAUNet will hold considerable implications for a large-scale agricultural parcel mapping in future.

5.3. Limitations

The process of extracting agricultural parcels from remote sensing images necessitates the fusion of parcel segmentation results with parcel

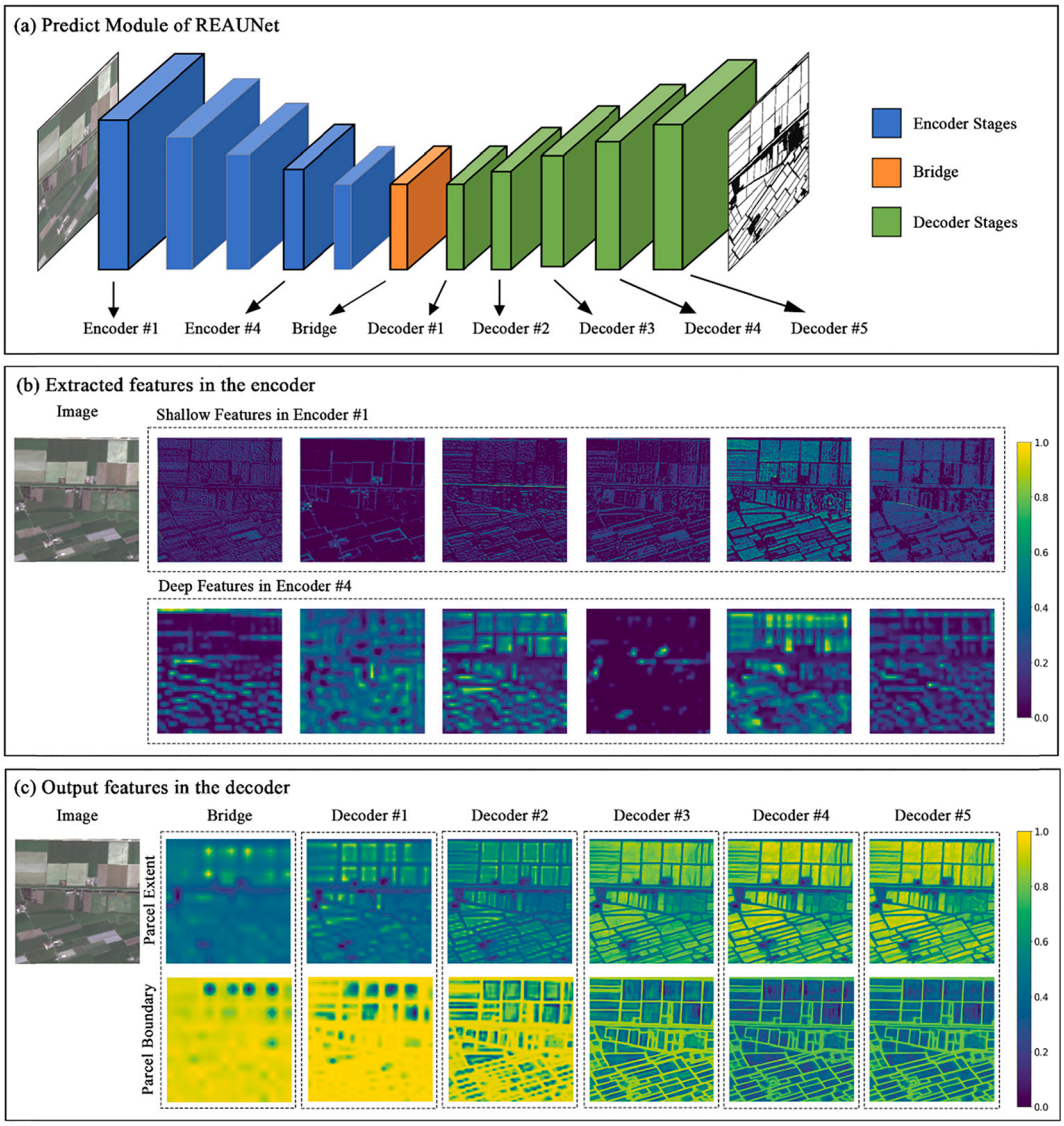

Fig. 7. Model feature maps in the encoder and decoder. (a) The predict module of REAUNet. (b) Extracted feature maps from shallow and deep layers in the encoder. (c) Output features of parcel extent and boundaries in the decoder.

Table 3
Accuracy assessment of different deep learning models for parcel delineation.

Method	NL						HT					
	OA	F1	IoU	GOC	GUC	GTC	OA	F1	IoU	GOC	GUC	GTC
U-Net	0.9299	0.9230	0.8305	0.1763	0.2614	0.2557	0.9391	0.9460	0.8979	0.1569	0.2631	0.2566
RCF	0.9130	0.9012	0.7961	0.2005	0.2989	0.2922	0.9372	0.9441	0.8946	0.1912	0.2315	0.2580
MPSPNet	0.9316	0.9289	0.8401	0.1422	0.2176	0.2384	0.9472	**0.9533**	0.9012	0.1897	**0.1908**	0.2309
ResUNet-a	0.9304	0.9252	0.8340	0.1931	0.2431	0.2459	0.9411	0.9478	0.9011	0.1503	0.2541	0.2483
BsiNet	0.9253	0.9195	0.8246	0.1680	0.2580	0.2578	0.9318	0.9395	0.8863	0.1954	0.2540	0.2761
FRNet	0.9314	0.9264	0.8359	0.1537	0.2388	0.2446	0.9440	0.9512	0.9076	0.2545	0.3033	0.3239
RAUNet	0.9180	0.9122	0.8132	0.2209	0.2592	0.2754	0.9458	0.9526	0.9035	0.2088	0.2224	0.2586
SEANet	0.9332	0.9304	**0.8425**	**0.1350**	0.2277	0.2355	0.9388	0.9466	0.8988	0.1598	0.2228	0.2314
REAUNet	**0.9473**	**0.9498**	0.8372	0.1363	**0.2148**	**0.2017**	**0.9481**	0.9454	**0.9096**	**0.1123**	0.2234	**0.2074**

edge results. We chose the simplest cutoff method to merge two model results, because Waldner and Diakogiannis (2020) found that it achieved similar results to the watershed approach. We observed that in the edge mask, the agricultural parcel edges were represented with varying probabilities, rendering the difficulty in selecting a suitable threshold. There is a trade-off between under-segmentation and over-segmentation errors wherein the choice of thresholds. How to use a better fusion method to merge the parcel extent mask and edge mask could be the key to further enhance the delineation accuracy.

We extracted refined agricultural parcels from model predictions through a post-processing workflow. However, in some instances, the edge probability maps output by the model exhibited lower probabilities due to occlusion or blurring, resulting in a complication for the threshold determination on the edge binarization. The generated lines

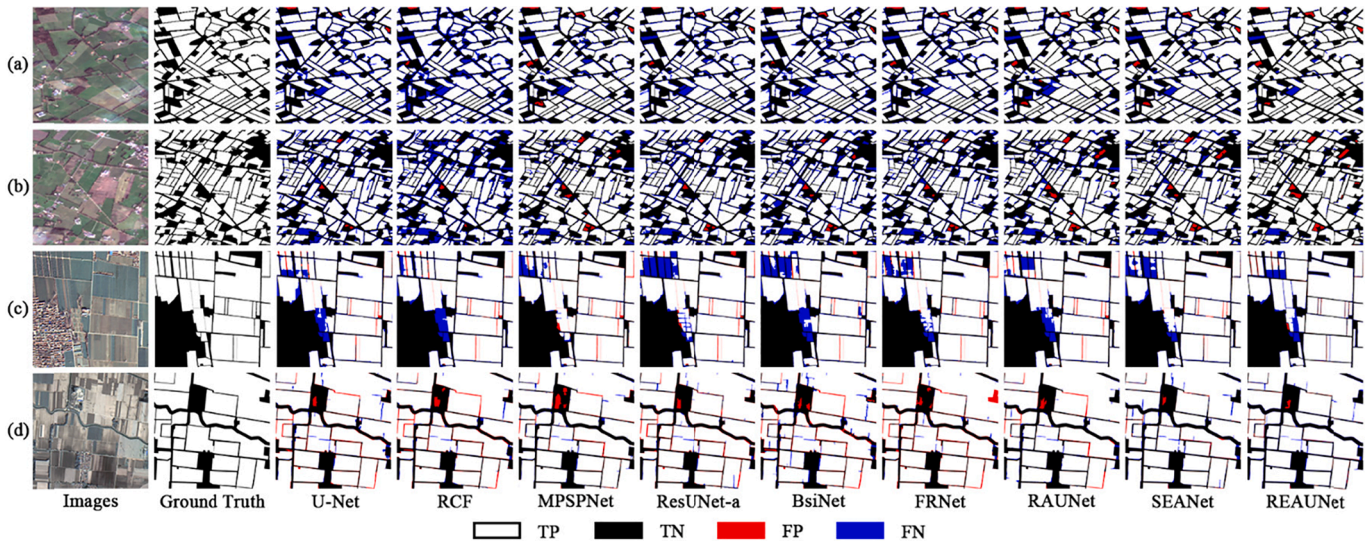

Fig. 8. Delineation results of REAUNet and other deep learning models in the comparative experiment. (a) and (b) represent area NL, while (c) and (d) represent area HT.

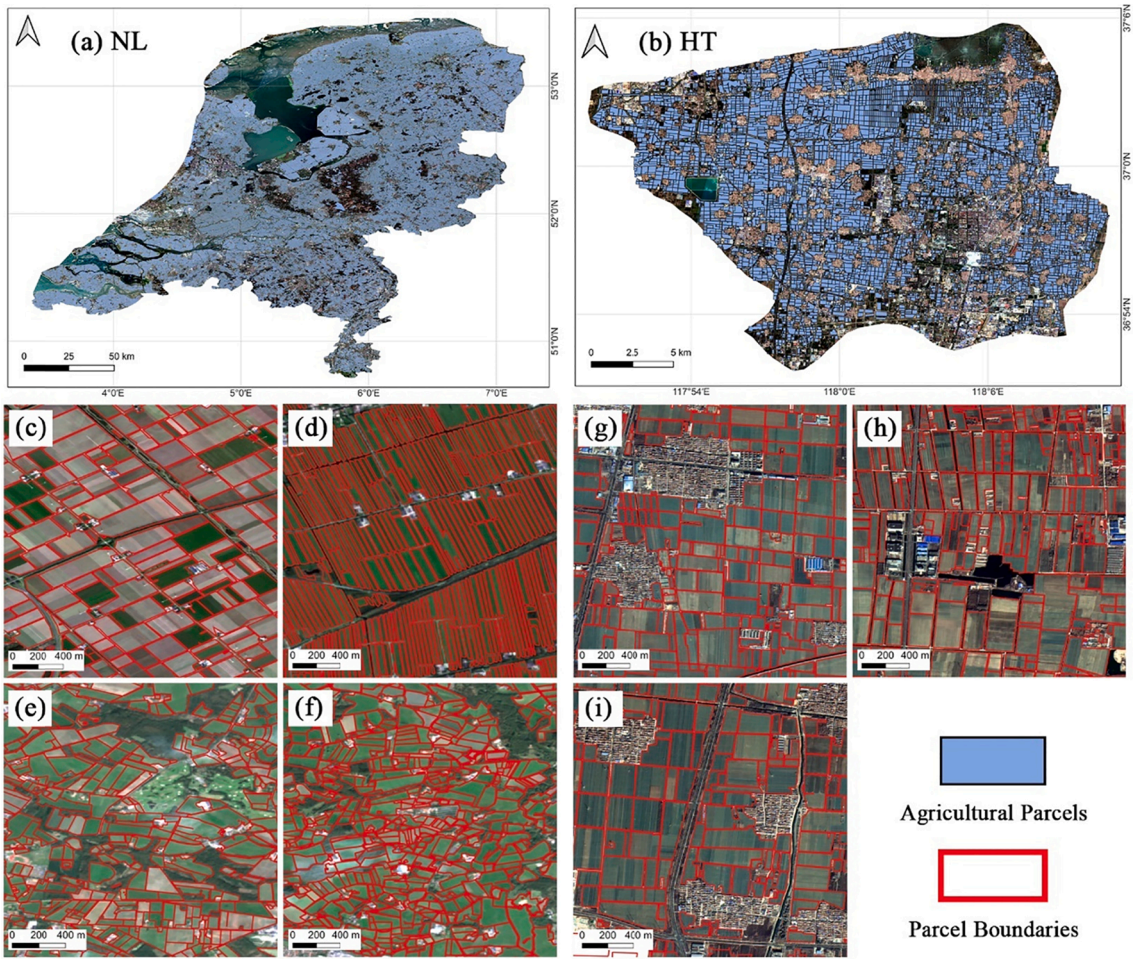

Fig. 9. The general delineation results of agricultural parcels and vectorized boundaries in (a) NL and (b) HT, and their subregions were shown in from (c) to (i): (c) and (i) shows regular parcels, (d) depicts narrow parcels, and (g) presents parcels of different sizes, (e) and (f) shows mountainous regions.

might end up being incomplete with breakpoints. Our current approach opts to directly remove short and isolated lines, which may advertently eliminate some incomplete but correct edges. In future research, the development of the breakpoint reconnection method is essential to form complete edges and mitigate the under-segmentation phenomenon in the delineation of agricultural parcels.

The biases in data labels may cause the loss of model accuracy. The open-source datasets may contain inconsistencies in parcel boundaries

Table 4
Accuracy assessment in the model transferability experiment. NT: no training. DT: direct training. TT: transfer training.

Study area	Method	OA	F1	IoU	GOC	GUC	GTC
DK	$REAUNet_{NT}$	0.7704	0.8113	0.6615	0.3925	0.4451	0.4565
	$REAUNet_{DT}$	0.8681	0.9028	0.7919	0.1571	0.2729	0.2599
	$REAUNet_{TT}$	**0.8887**	**0.9195**	**0.8181**	**0.1424**	**0.2460**	**0.2343**
TR	$REAUNet_{NT}$	0.7134	0.7794	0.6185	0.4961	0.6380	0.5679
	$REAUNet_{DT}$	0.8979	0.9248	0.8602	0.1403	0.1942	0.1891
	$REAUNet_{TT}$	**0.9001**	**0.9270**	**0.8640**	**0.1318**	**0.1823**	**0.1781**

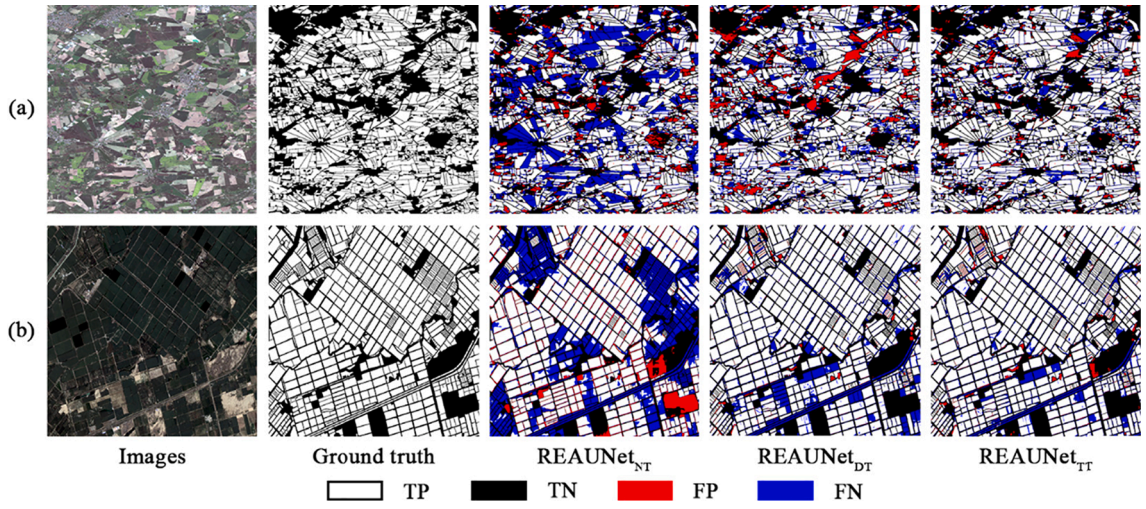

Fig. 10. Model results in the model transferability experiment. (a) represents area DK and (b) represents area TR.

due to temporal discrepancies, leading to inaccuracies in defining the exact shape and size of agricultural parcels. Additionally, there may be variability in whether parcel ridges are marked, introducing further inconsistencies. These deviations can affect the training process, making it challenging for the model to find the correct fitting direction and thereby reducing accuracy. To mitigate the influence of data errors, future efforts can implement collaborative labeling processes involving multiple individuals to minimize accidental human errors. Employing consensus-based labels can enhance the reliability of identified agricultural parcels. Moreover, integrating supplementary data sources for cross-validation purposes may bolster the accuracy and robustness of the delineated agricultural parcels.

We evaluated the model performance in new regions through a transfer experiment and observed that even a small number of local samples could markedly boost model accuracy. Given the labor-intensive sample labeling, there is always an expectation to achieve a high extraction accuracy with only minimal local samples. However, when deploying the model to a new area, determining the optimal number of local samples for transfer training remains a significant challenge, which could be a limitation for our study. Our future research will focus on exploring the optimal number of samples for transfer learning to address this gap and enhance the robustness and generalizability of our model. Strategically selecting local samples will significantly facilitate large-scale mapping and reduce the time and resources required for sample preparation and model training.

6. Conclusion

This study developed a novel deep learning model, REAUNet, to delineate agricultural parcels from high-resolution remote sensing images. Four model innovations were ensembled to improve edge-awareness, including the deep supervision, the edge detection block, the dual attention block and the refine module. Sentinel-2 and GF-2 imagery at different resolutions across four regions were utilized to assess the effectiveness of parcel delineation. To optimize the geometry and shape of extracted parcels, we formulated a post-processing workflow utilizing the Douglas-Peucker algorithm and an over-sharp angle detection and elimination algorithm. Comparative experiments revealed that our proposed model surpassed representative agricultural parcel extraction CNN models in both thematic and geometric accuracy. Finally, to facilitate the successful delineation of agricultural parcels in unknown regions using the proposed deep learning model, we tested and confirmed the capability of the new model for transferring pre-models to a new region with a small proportion of local samples. This opened up the possibility of mapping the distribution of agricultural parcels on a large scale. In conclusion, our proposed algorithm successfully delineated the agricultural parcels from remote sensing images. Enhancing the edge-awareness of the model significantly improved its accuracy in delineating agricultural parcels, particularly on the parcel edges. The method also demonstrated good generalization results in different regions with limited local samples, offering a new approach for the automatic mapping of agricultural parcels at a large spatial scale.

CRediT authorship contribution statement

Rui Lu: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yingfan Zhang:** Writing – review & editing, Methodology, Investigation. **Qiting Huang:** Supervision, Data curation. **Penghao Zeng:** Writing – review & editing. **Zhou Shi:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Su Ye:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgements

The authors are very grateful for the financial support provided by National Key Research and Development Program (2022YFB3903503), and by the Fundamental Research Funds for the Central Universities (226-202400052). The authors would also like to thank the editors and anonymous reviewers of the journal for their thorough review and insightful comments that have helped improving the quality of the paper.

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